

Confidence Intervals for Proportions and Means

Module 2, Part 2 (October 16)

QBA 5330

Where We Left Off

We have a point estimate and a standard error for the pilot launch data.

Missing: a defensible range to bring to leadership.

point estimate $\pm$ critical value $\times$ standard error

Confidence Interval for the Conversion Rate

,, 95% confidence

```
1 p_hat <- 0.15
2 n <- 200
3 se <- sqrt(p_hat * (1 - p_hat) / n)
4 z <- qnorm(0.975)
5
6 c(lower = p_hat - z * se, upper = p_hat + z * se)

lower      upper
0.1005133 0.1994867
```

“We’re 95% confident the true national conversion rate is between about 10.1% and 19.9%.”

Confidence Interval for the Order Value

,,, 95% confidence

```
1 x_bar <- 71.50
2 s <- 42
3 n <- 25
4 se <- s / sqrt(n)
5 t <- qt(0.975, df = n - 1)
6
7 c(lower = x_bar - t * se, upper = x_bar + t * se)
```

```
lower    upper
54.16325  88.83675
```

Your Turn: Does the Pilot Clear the Bar? (15 minutes)

Leadership says the launch is worth pursuing only if:

- True national conversion rate is **at least 10%**
- True national average order value is **at least \$60**
- **Check both intervals against these thresholds. Report out.**

What Your Group Should Find

- Conversion: – barely clears 10%
- Order value: – **includes** values below \$60

Before the Break

- Confidence intervals give us a **range** of plausible values.
- But leadership asked a **yes/no question**: does the average order value exceed \$60?
- Formalizing that yes/no question is exactly what a **hypothesis test** does.